

Distribution Function or Cumulative Distribution Function

The distribution of a r.v. X is defined as

$$F_X(x) = P(X \leq x) \quad ; \quad -\infty < x < \infty$$

Properties of DF: ① If $a < b$, then $P(a < X \leq b) = F(b) - F(a)$

Proof: The events $a < X \leq b$ and $X \leq a$ are disjoint, hence their union is $X \leq b$

$$\text{i.e. } (a < X \leq b) \cup (X \leq a) = X \leq b.$$

$$P(a < X \leq b) + P(X \leq a) = P(X \leq b)$$

$$\Rightarrow P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$

Cor 1

$$\begin{aligned} P(a \leq X \leq b) &= P(X=a \cup (a < X \leq b)) \\ &= P(X=a) + P(a < X \leq b) \\ &= P(X=a) + [F(b) - F(a)] \end{aligned}$$

Similarly

$$\begin{aligned} P(a < X < b) &= P(a < X \leq b) - P(X=b) \\ &= F(b) - F(a) - P(X=b) \end{aligned}$$

$$\begin{aligned} P(a \leq X < b) &= P(a < X < b) + P(X=a) \\ &= F(b) - F(a) - P(X=b) + P(X=a) \end{aligned}$$

Remark: ① When $P(X=a)=0$ and $P(X=b)=0$, all four events $a \leq X \leq b$, $a < X < b$, $a \leq X < b$ and $a < X \leq b$ have the same probability $F(b) - F(a)$

② (i) $0 \leq F(x) \leq 1$ (ii) For $x < y$ $F(x) \leq F(y)$

③ $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$

Prob Mass Function (pmf): If X is a discrete r.v. with distinct values $x_1, x_2, \dots, x_n, \dots$ then the function $p(x)$ defined as

$$p(x) = \begin{cases} P(X=x) = p(x_i), & \text{if } x = x_i \\ 0, & \text{if } x \neq x_i; i=1,2,\dots \end{cases}$$

is called pmf if $p(x_i) \geq 0 \forall i$ and $\sum_{i=1}^{\infty} p(x_i) = 1$.

Discrete Distribution Function

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$$F_X(x) = P(X \leq x) = \sum_{x \leq x} p(x)$$

Ex.

X	x_1	x_2	x_3	x_4	x_5	x_6
$p(x)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	$p(x_4)$	$p(x_5)$	$p(x_6)$

$$F_X(x_4) = p(x_1) + p(x_2) + p(x_3) + p(x_4)$$

Ex. A r.v. X has the following prob function

x	0	1	2	3	4	5	6	7
$p(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

① Find K (ii) Evaluate $P(X < 6)$ (iii) $P(X > 6)$

iv) $P(0 < X < 5)$ (v) If $P(X \leq a) > \frac{1}{2}$ find the minimum value of a , and vi) determine the distⁿ function of X .

i) Since $\sum_{x=0}^7 p(x) = 1$, $0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$

$$\Rightarrow 10K^2 + 9K - 1 = 0 \Rightarrow (10K - 1)(K + 1) = 0$$

$$\Rightarrow K = \frac{1}{10} \text{ or } K = -1 \text{ but } p(x) \text{ can't be -ve}$$

$$\text{hence } K = \frac{1}{10}$$

ii) $P(X < 6) = p(0) + p(1) + p(2) + p(3) + p(4) + p(5)$
 $= \frac{81}{100}$

iii) $P(X > 6) = 1 - P(X < 6) = \frac{19}{100}$

iv) $P(0 < X < 5) = p(1) + p(2) + p(3) + p(4) = \frac{4}{5}$

v) $P(X \leq a) > \frac{1}{2}$ By trial we get $a = 4$

vi) distⁿ function of X

x	0	1	2	3	4	5	6	7
$p(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$
$F_X(x)$ $= P(X \leq x)$	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{4}{5}$	$\frac{81}{100}$	$\frac{83}{100}$	1

Ex. If $p(x) = \frac{x}{15}$; $x = 1, 2, 3, 4, 5$
 $= 0$; elsewhere

Find i) $P(X=1 \text{ or } 2)$ ii) $P\left\{\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right\}$

i) $P(X=1 \text{ or } 2) = P(X=1) + P(X=2) = \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$

ii) $P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{P\left(\frac{1}{2} < X < \frac{5}{2} \cap X > 1\right)}{P(X > 1)}$
 $= \frac{P(X=1 \text{ or } 2 \cap X > 1)}{P(X > 1)} = \frac{P(X=2)}{1 - P(X=1)}$
 $= \frac{2/15}{1 - \frac{1}{15}} = \frac{2}{15} \times \frac{15}{14} = \frac{1}{7}$

Continuous Random variable : can take all possible values between certain limits.

Prob Density Function

$P(x \leq X \leq x+dx) = f(x)dx$

Defn: p.d.f. $f_X(x)$ of the r.v. X is defined as

$$f_X(x) = \lim_{dx \rightarrow 0} \frac{P(x \leq X \leq x+dx)}{dx}$$

$$= \lim_{dx \rightarrow 0} \frac{F(x+dx) - F(x)}{dx} = \frac{d}{dx} F(x)$$

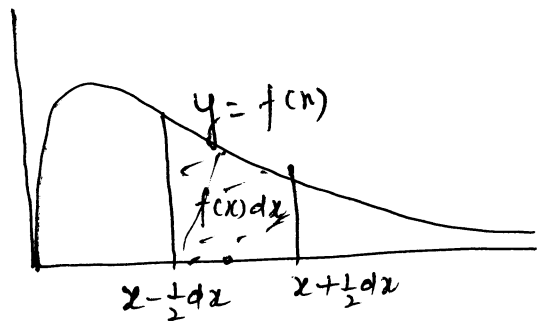
$f_X(x) = \frac{d}{dx} F(x)$ — relation between distⁿ function pdf & DF.

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$f(x)$ is pdf iff ① $f(x) \geq 0 \forall x \in (-\infty, \infty)$

② $\int_{-\infty}^{\infty} f(x) dx = 1$

Distribution Function $F_X(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$



Results :

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if $X \sim f(x)$; $a < x < b$

$$AM = \int_a^b x f(x) dx$$

$$HM = \frac{1}{H} = \int_a^b \frac{1}{x} f(x) dx$$

$$GM \quad \log G = \int_a^b \log x f(x) dx$$

$$\mu_r' = \int_a^b x^r f(x) dx$$

$$\mu_r = \int_a^b (x - \text{mean})^r f(x) dx$$

Median $\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$

Thus solving $\int_a^M f(x) dx = \frac{1}{2}$ or $\int_M^b f(x) dx = \frac{1}{2}$

M.D $M.D_x = \int_a^b |x - \text{mean}| f(x) dx$

$$M.D_A = \int_a^b |x - A| f(x) dx$$

Quartile and Deciles

$$\int_a^{Q_1} f(x) dx = \frac{1}{4} \quad \text{and} \quad \int_a^{Q_3} f(x) dx = \frac{3}{4}$$

i^{th} Deciles $D_i \quad \int_a^{D_i} f(x) dx = \frac{i}{10} \quad (i = 1, 2, \dots, 9)$

Mode : It is the value at which $f(x)$ is maximum
thus the solⁿ of $f'(x) = 0$ and $f''(x) < 0$,
provided it lies in $[a, b]$.

Ex. The diameter of an electric cable, say x , is assumed to be a cont. r.v. with pdf $f(x) = 6x(1-x)$, $0 \leq x \leq 1$, (1) check $f(x)$ is pdf
(ii) Find b if $P(X < b) = P(X > b)$.

Two-Dimensional Random variables

It is possible to define more than one random variable on the same sample space. Ex. Height + weight of every person in certain educational institution.

Defⁿ: Let X and Y be two random variables defined on the same sample space S , then the function (X, Y) that assigns a point in $R^2 (= R \times R)$, is called a two-dimensional random variable.

Let $A = [a < X \leq b]$, $B = [c < Y \leq d]$ be two events, then the event $[a < X \leq b, c < Y \leq d] = \{a < X \leq b\} \cap \{c < Y \leq d\} = A \cap B$

$$\therefore P(a < X \leq b, c < Y \leq d) = P(A \cap B)$$

$$P_{XY}(x, y) = P[(X, Y) \in E]$$

$$p_{ij} = P(X = x_i \cap Y = y_j) = p(x_i, y_j)$$

Y	x_1	x_2	x_3	$\dots$	x_j	$\dots$	x_m	Total
x_1	p_{11}	p_{12}	p_{13}		p_{1j}		p_{1m}	$p_{1\cdot}$
x_2	p_{21}	p_{22}	p_{23}		p_{2j}		p_{2m}	$p_{2\cdot}$
x_3	p_{31}	p_{32}	p_{33}		p_{3j}		p_{3m}	$p_{3\cdot}$
$\vdots$								
x_i	p_{i1}	p_{i2}	p_{i3}		p_{ij}	$\dots$	p_{im}	$p_{i\cdot}$
$\vdots$								
x_n	p_{n1}	p_{n2}	p_{n3}		p_{nj}	$\dots$	p_{nm}	$p_{n\cdot}$
Total	$p_{\cdot 1}$	$p_{\cdot 2}$	$p_{\cdot 3}$		$p_{\cdot j}$		$p_{\cdot m}$	1

Two dimensional or Joint Prob Mass Function

Defⁿ: If (X, Y) is a two-dimensional discrete r.v., then the joint discrete function of X, Y , also called the joint prob mass function of X, Y denoted by $p_{XY}(x, y)$ is defined as

$$P_{XY}(x_i, y_j) = P(X = x_i, Y = y_j) \text{ for a value of } (x_i, y_j) \text{ of } (X, Y)$$

$$= 0, \text{ otherwise.}$$

$$\sum_i \sum_j p_{XY}(x_i, y_j) = 1 \quad \& \quad p_{XY}(x_i, y_j) \geq 0 \quad \forall i, j$$

Marginal Prob Function (or Marginal Prob Mass Function)

$$p_X(x_i) = \sum_{j=1}^m p(x_i, y_j) \quad \text{and} \quad p_Y(y_j) = \sum_{i=1}^n p(x_i, y_j) = p_{\cdot j} \\ = p_{\cdot j}$$

Conditional Prob Function

$$p_{X|Y}(x/y) = \frac{P(X=x, Y=y)}{P(Y=y)} \quad \text{provided } P(Y=y) \neq 0 \\ = \frac{p(x, y)}{p_{\cdot j}}$$

$$\text{Similarly } p_{Y|X}(y/x) = \frac{P(X=x, Y=y)}{P(X=x)} = \frac{p(x, y)}{p(x)}; \quad p(x) \neq 0$$

Two-dimensional Distribution Function

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

Marginal Distⁿ Function

$$F_X(x) = P(X \leq x) = P(X \leq x, Y < \infty) = \lim_{y \rightarrow \infty} F_{XY}(x, y) \\ = F_{XY}(x, \infty)$$

$$\text{Similarly } F_Y(y) = P(Y \leq y) = P(X < \infty, Y \leq y) = \lim_{x \rightarrow \infty} F_{XY}(x, y) \\ = F_{XY}(\infty, y)$$

$$\Rightarrow F_X(x) = \sum_y P(X \leq x, Y=y) \quad \text{and} \quad F_Y(y) = \sum_x P(X=x, Y \leq y)$$

Similarly for continuous r.v

$$F_X(x) = \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dy \right\} dx$$

$$F_Y(y) = \int_{-\infty}^y \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dx \right\} dy$$

The marginal prob function of X and Y are given respectively as follows:

$$f_X(x) = \begin{cases} \sum_y p(x, y) & (\text{for discrete r.v}) \\ \int_{-\infty}^{\infty} f(x, y) dy & (\text{for cont. r.v.}) \end{cases}$$

and
$$f_Y(y) = \begin{cases} \sum_x p(x,y) & \text{for discrete r.v.} \\ \int_{-\infty}^{\infty} f(x,y) dx & \text{for continu. r.v.} \end{cases}$$

The marginal density functions of x and y can be obtained in the following manner also:

$$f_X(x) = \frac{\partial F_X(x)}{\partial x} = \int_{-\infty}^{\infty} f_{XY}(x,y) dy$$

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = \int_{-\infty}^{\infty} f_{XY}(x,y) dx$$

⇒ The Conditional Distribution Function and Conditional Probability density function:

We know $F_{XY}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x P(A|X=x) dF_X(x)$

$$F_{Y|X}(y|x) = P(Y \leq y | X=x) = P(A | X=x) \quad \text{where } A \text{ is } Y \leq y$$

where A be the event $Y \leq y$

$$F_{XY}(x,y) = \int_{-\infty}^x F_{Y|X}(y|x) dF_X(x)$$

$$= \int_{-\infty}^x F_{Y|X}(y|x) f_X(x) dx$$

Similarly
$$F_{XY}(x,y) = \int_{-\infty}^y F_{X|Y}(x|y) dF_Y(y)$$

$$= \int_{-\infty}^y F_{X|Y}(x|y) f_Y(y) dy$$

$$\left. \begin{aligned} F(A|B) &= \frac{P(A \cap B)}{P(B)} \\ P(A \cap B) &= P(A|B) P(B) \end{aligned} \right\}$$

$$\left. \begin{aligned} \frac{dF_X(x)}{dx} &= f_X(x) \\ dF_X(x) &= f_X(x) dx \end{aligned} \right\}$$

For the

$$\left\{ \begin{aligned} f_{Y|X}(y|x) &= \frac{\partial}{\partial y} F_{Y|X}(y|x) \\ f_{X|Y}(x|y) &= \frac{\partial}{\partial x} F_{X|Y}(x|y) \end{aligned} \right\}$$

Stochastic Independence :

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$$If \quad f_{xy}(x, y) = f_x(x) g_y(y)$$

Ex. The joint prob distribution of two random variables X and Y given below

X \ Y	1	2	3	4	Total
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
Total	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Find (i) The marginal distributions of X and Y
 (ii) Conditional distⁿ of X given the value of Y=1 and that of Y given the value of X=2.

Solⁿ: The marginal distⁿ of X is defined as $P(X=x) = \sum_y P(X=x, Y=y)$

X	1	2	3	4
P(X)	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

Similarly

Y	1	2	3	4
P(Y)	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

$$As \quad P(Y=y) = \sum_x P(X=x, Y=y)$$

$$P(X=x|Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$\text{hence } P(X=1|Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{\frac{4}{36}}{\frac{11}{36}} = \frac{4}{11}$$

$$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{\frac{1}{36}}{\frac{11}{36}} = \frac{1}{11}$$

$$P(X=3|Y=1) = \frac{P(X=3, Y=1)}{P(Y=1)} = \frac{\frac{5}{36}}{\frac{11}{36}} = \frac{5}{11}$$

$$P(X=4|Y=1) = \frac{P(X=4, Y=1)}{P(Y=1)} = \frac{\frac{1}{36}}{\frac{11}{36}} = \frac{1}{11}$$

Hence the Conditional distⁿ of X given Y=1 is

$$P(X=x|Y=1) \rightarrow$$

X	1	2	3	4
P(X=x Y=1)	$\frac{4}{11}$	$\frac{1}{11}$	$\frac{5}{11}$	$\frac{1}{11}$

Similarly we can obtain conditional distⁿ of Y given X=2 as

Y	1	2	3	4
P(Y=y X=2)	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{9}$